

POLITECNICO
MILANO 1863

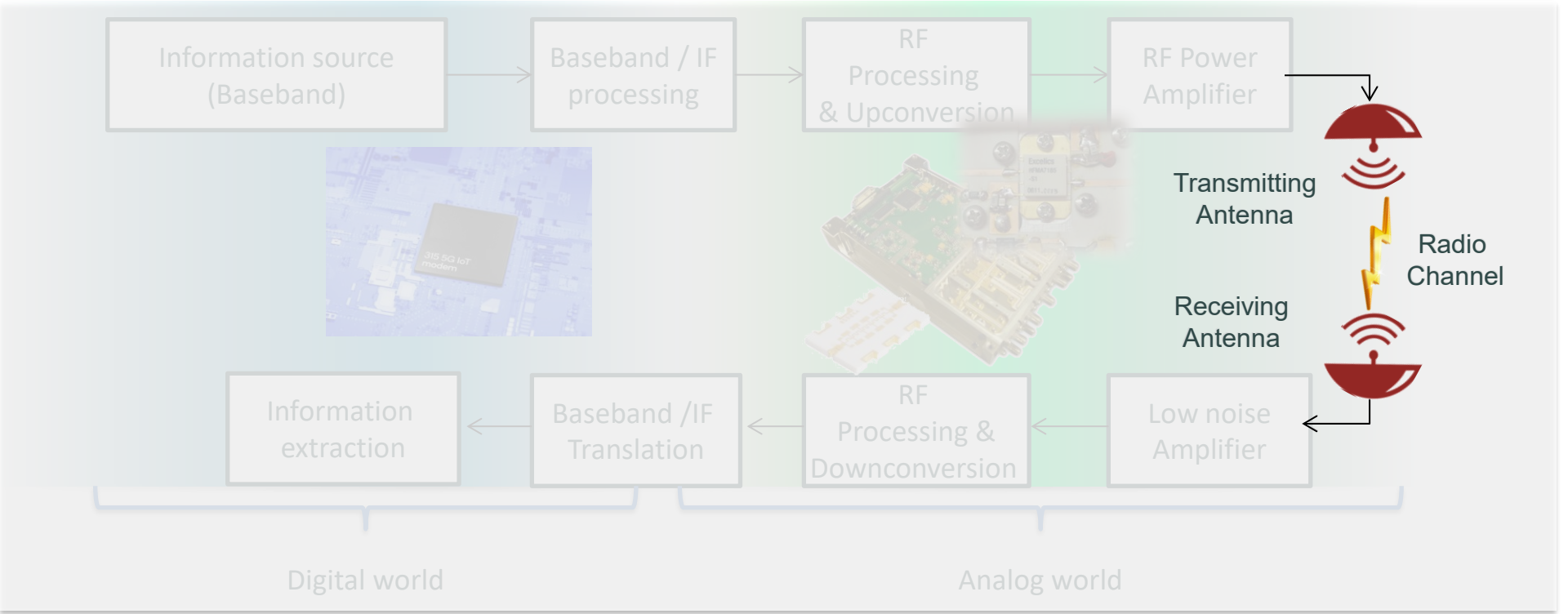
DIPARTIMENTO DI ELETTRONICA
INFORMAZIONE E BIOINGEGNERIA

APPLIED ELECTROMAGNETICS @POLIMI

RF Systems

Antennas
Link Budget
Noise
Data Rate
Radar and Satellite Communications

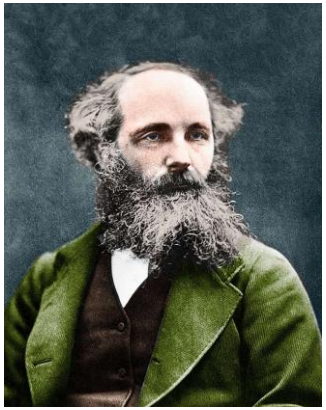
RF Frontend

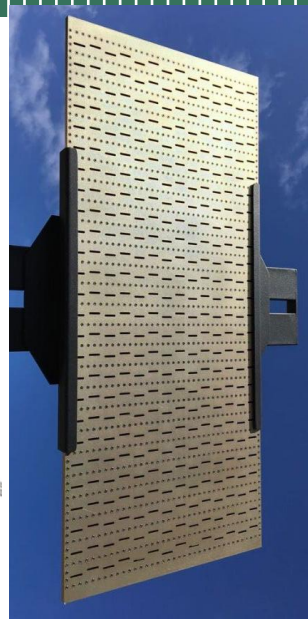

Information source (Baseband)
Baseband / IF processing
RF Processing & Upconversion
RF Power Amplifier
Transmitting Antenna
Radio Channel
Receiving Antenna
Low noise Amplifier
RF Processing & Downconversion
Baseband /IF Translation
Information extraction
Digital world
Analog world

Radiated Transmission

$$\begin{cases} \nabla \times \tilde{E} = -j\omega\tilde{B} \\ \nabla \times \tilde{H} = j\omega\tilde{D} + \tilde{J} \\ \nabla \bullet \tilde{D} = \rho_v \\ \nabla \bullet \tilde{B} = 0 \end{cases}$$

Maxwell Equations (1865)















Radiated Transmission

The electromagnetic field:

can be “guided” by transmission lines

“radiated” in the space by suitable transducers: **antennas**

The propagation in the free space is governed by Maxwell equations.

Sufficiently far (in *far field*), the radiated wave is spherical and locally approximated as a plane wave (E and H vectors in a plane orthogonal to the propagation direction and orthogonal to each other)

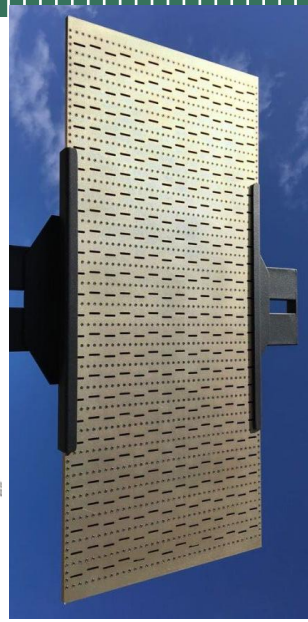












Let there be light

MAXWELL EQUATIONS
DIFFERENTIAL FORM

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$ GAUSS'S LAW FOR ELECTRICITY

$$\nabla \cdot \mathbf{B} = 0$$ GAUSS'S LAW FOR MAGNETISM

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$ FARADAY'S LAW OF INDUCTION

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}$$ AMPÈRE–MAXWELL LAW

MAXWELL EQUATIONS
INTEGRAL FORM

$$\oint_S \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{enc}}{\varepsilon_0}$$ GAUSS'S LAW FOR ELECTRICITY

$$\oint_S \mathbf{B} \cdot d\mathbf{A} = 0$$ GAUSS'S LAW FOR MAGNETISM

$$\oint_C \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi_B}{dt}$$ FARADAY'S LAW OF INDUCTION

$$\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc} + \mu_0 \varepsilon_0 \frac{d\Phi_E}{dt}$$ AMPÈRE–MAXWELL LAW

VALID FOR EVERY ELECTRIC/MAGNETIC/ELECTROMAGNETIC CIRCUIT OR EQUIPMENT FROM DC TO INFINITE FREQUENCY (NEGLECTING QUANTUM EFFECTS AT MICROSCOPIC LEVEL)

Phasor representation of a plane wave propagating along z

Linear polarization:

(i.e. horizontal or vertical)

→Tx and Rx antennas must be aligned along the same polarization!

Circular polarization:

(i.e. right- or left-handed)

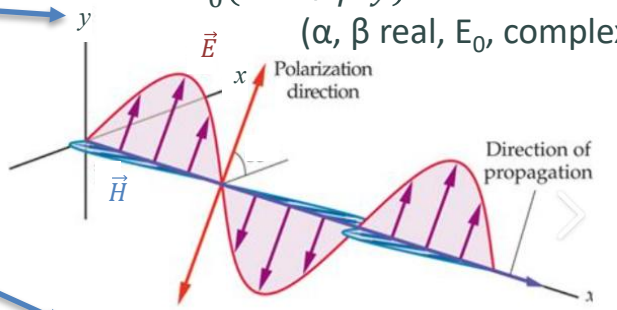
→Tx and Rx antennas must use the same polarization diriection!
Typically used in satellite links

Impedance: regardless of polarization, The magnetic field *H* is orthogonal to *E* and to the propagation direction and the ratio E/H is the wave impedance Z_W or η:

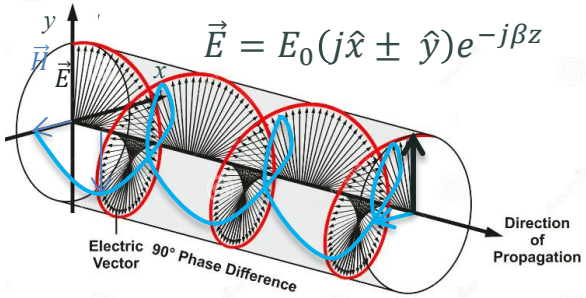
$$\frac{E}{H} = Z_W \qquad \text{With:} \quad Z_W = \frac{377\Omega}{\sqrt{\epsilon_r}}$$

$$\vec{E} = E_0(\alpha\,\hat{x} + \beta\,\hat{y})e^{-j\beta z}$$

(α, β real, E_0, complex)

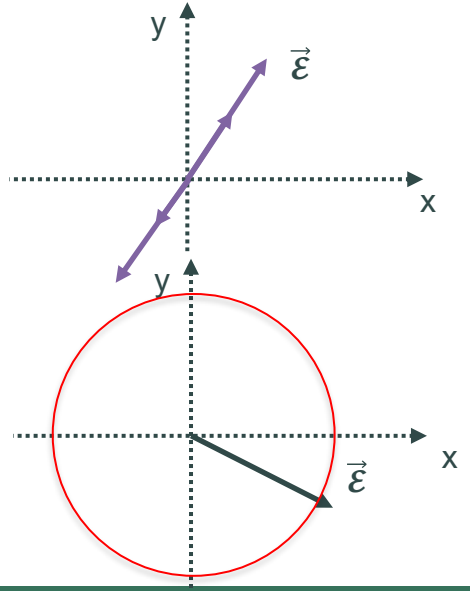


$$\vec{E} = E_0(j\hat{x} \pm \hat{y})e^{-j\beta z}$$



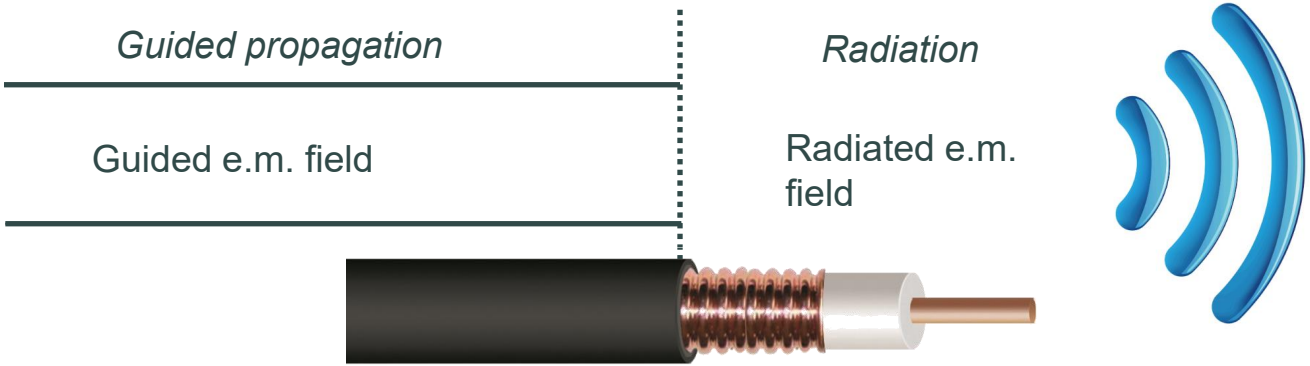
In air or vacuum, ε_r=1 → Z_W=377 Ω

Electric field in time:



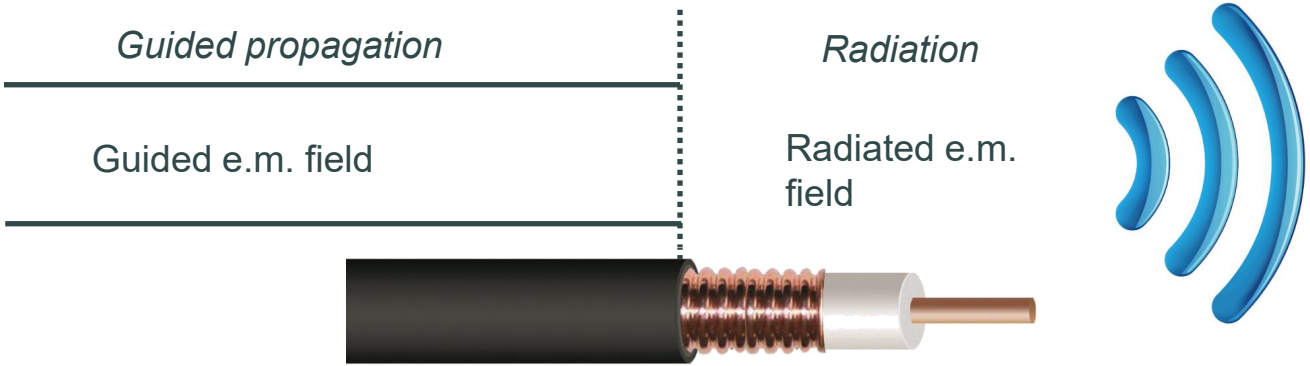
Generation of a plane wave in the space

For instance: a transmission line terminating abruptly (truncated cable)



Generation of a plane wave in the space

For instance: a transmission line terminating abruptly (truncated cable)



- Truncated cable: very inefficient way to produce radiation (most of the power is reflected back to the transmission line)
- A true **Antenna** is a sort of matching structure that makes as efficient as possible the transition from guided to radiated propagation

Types of Antennas

Many different types of antennas

Choice depends on: frequency, size, cost,…

In this course we will consider only the properties of the antennas impacting on the radio link.

Antennas are passive and reciprocal → They behave the same in Tx and Rx

Two main categories:

- Omni-directional: radiate uniformly around them
- Directional: privilege a specific direction







Other courses dedicated entirely to antennas: “Antennas”, 2nd semester!

E.M. Field generated by the antenna

The generated field includes two contributions:

- the Near Field varying as $1/R^2$
- the Far Field varying as $1/R$

"far field": beyond the Fraunhofer distance of that antenna: $d_F=2(\text{MaxAntennaDimension})^2/\lambda$

At a point (R,θ,ϕ) in "far field" region, near field is practically negligible, and the far field is approx. a plane wave.

The real power density (magnitude of Poynting Vector) is defined as:



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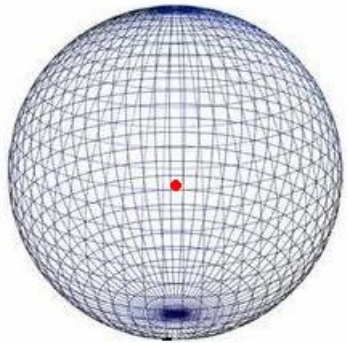
The real power density (magnitude of Poynting Vector) is defined as:

$$\vec{S}_R = \frac{1}{2} Re\left(\vec{E} \times \vec{H}^*\right) \propto \frac{1}{R^2}\hat{\imath}_r$$

If the power P_{rad} is radiated by an isotropic antenna (uniformly spreads in every direction)

$$S_R = \frac{P_{rad}}{4\pi R^2}$$

[surface of sphere=4 π R^2]





Directional properties of the radiated power

- An isotropic radiator does not exist.
- Any antenna has one or more “preferred” directions:

  → Radiated power density from an antenna depends on the direction
- For a given antenna, its *directivity* toward a given direction (θ, ϕ) is defined with respect to an ideal isotropic radiator radiating the same overall total power P_{rad}, both observed at arbitrary far field distance R:

$$D(\theta,\phi) = \frac{\text{Radiated Power Density by the antenna toward } \theta,\phi}{\text{Power Density radiated by an isotropic radiator}} = \frac{S_R(R,\theta,\phi)}{\frac{P_{rad}}{4\pi R^2}}$$ [does not depend on R]

- The radiated power density at a point (R, θ , φ) can then be expressed as:

$$S_R(R,\theta,\phi) = \frac{P_{rad}}{4\pi R^2} D(\theta,\phi)$$



Meaning of D_M

The directions $(\theta_{\max}, \phi_{\max})$ where *D* is maximum is/are the directions of maximum radiation.

$$D_M = \max(D(\theta,\varphi)) = D(\theta_{max},\varphi_{max})$$

The *directivity function* (≤1) is *D* normalized to D_M:

$$f(\theta,\phi) = \frac{D(\theta,\phi)}{D_M} \quad \Rightarrow \quad S_R(R,\theta,\phi) = \frac{P_{rad}}{4\pi R^2} D_M f(\theta,\phi)$$

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The *directivity function* (≤1) is D normalized to D_M:

$$f(\theta,\phi) = \frac{D(\theta,\phi)}{D_M} \quad \Rightarrow \quad S_R(R,\theta,\phi) = \frac{P_{rad}}{4\pi R^2} D_M f(\theta,\phi)$$



From the definition of D, D_M is the ratio of power density in the max radiation direction and the power density of an ideal isotropic radiator at same overall radiated power and distance R:

$$D_M = \frac{S_R(R,\theta_{max},\phi_{max})}{\frac{P_{rad}}{4\pi R^2}}$$

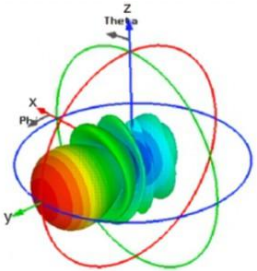
D_M does not depend on distance R!

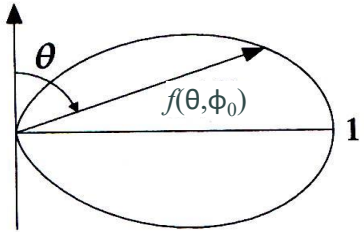
Max Directivity is often in log units: $D_{M\,\mathrm{dBi}} = 10\mathrm{Log}(D_M)$ [dBi] "dB with respect to an isotropic radiator"

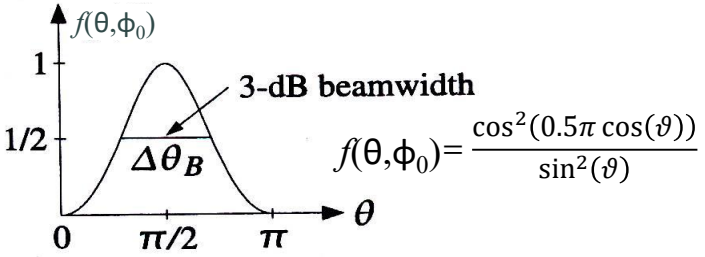
Plots of directivity function

The directivity function can be displayed either in polar coordinates (R, θ for fixed ϕ) or on a scalar plot (f vs θ or ϕ at specific "cuts"):

For instance:







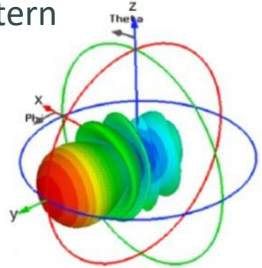
The parameter $\Delta\theta_B$ is called *half-power beamwidth* (HPBW) and is defined by the two values of θ where f is halved with respect to the direction of maximum

Directivity Diagram of real antennas

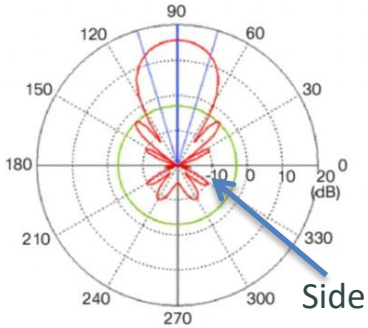
Directional pattern



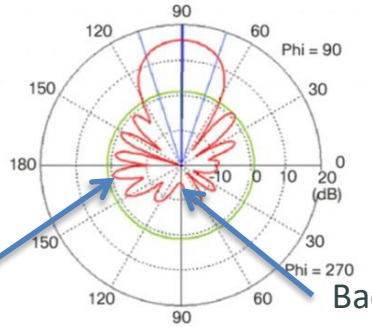
(a) Yagi Antenna Model



(b) Yagi Antenna 3D Radiation Pattern



(c) Yagi Antenna Azimuth Plane Pattern



(d) Yagi Antenna Elevation Plane Pattern

Side lobes

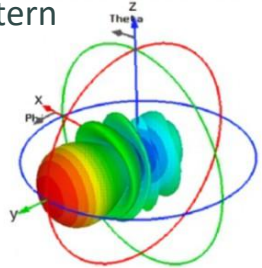
Back radiation

Directivity Diagram of real antennas

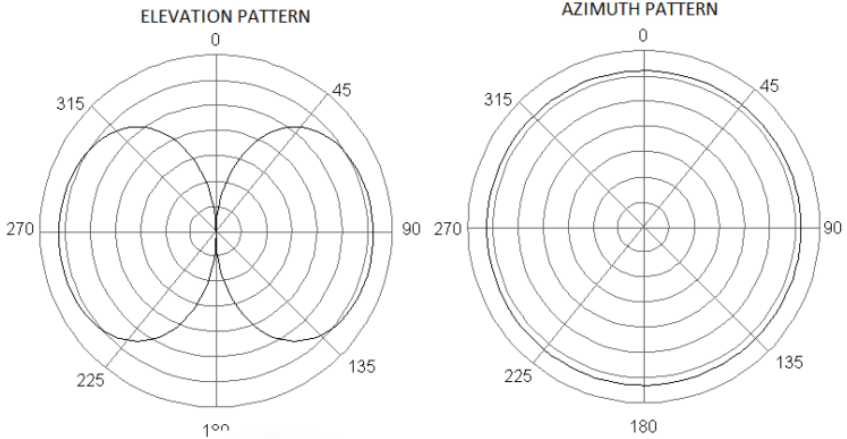
Directional pattern

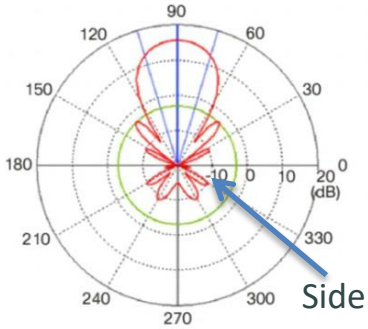


(a) Yagi Antenna Model

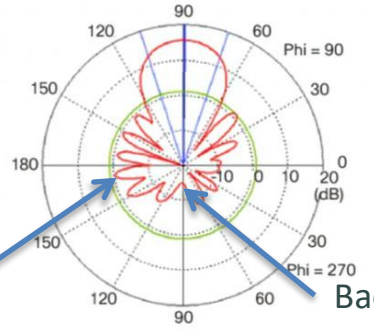


(b) Yagi Antenna 3D Radiation Pattern





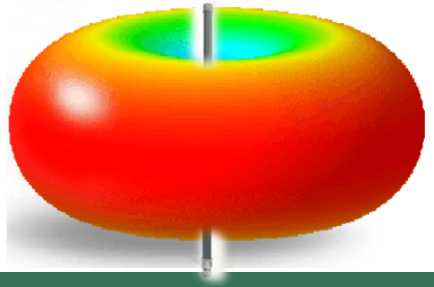
(c) Yagi Antenna Azimuth Plane Pattern



(d) Yagi Antenna Elevation Plane Pattern

Side lobes

Back radiation



Omni-directional pattern in azimuth

Evaluation of D from directivity function

From the formula of power density at distance R:

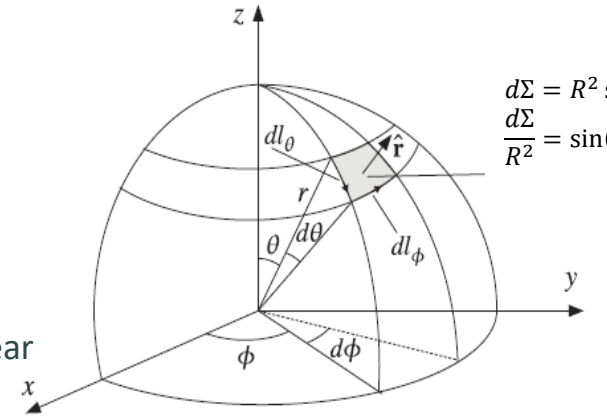
$$P_{rad} = \iint_{\Sigma} S_R(R,\vartheta,\varphi)d\Sigma \quad = \frac{P_{rad}D_M}{4\pi R^2}\iint_{\Sigma} f(\vartheta,\varphi)\,d\Sigma$$

Σ=Spherical surface with radius R.

→ D_M can be determined from $f(\theta,\phi)$!

$$D_M = \frac{4\pi}{\int_0^{\pi}\int_0^{2\pi} f(\theta,\varphi)\sin(\theta)\,d\varphi d\theta}$$

[D_M in linear scale]

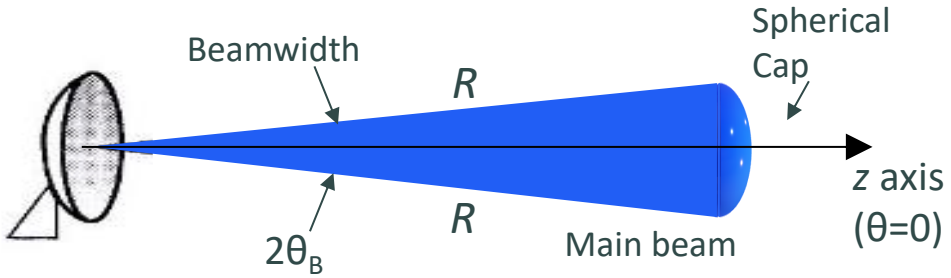


$$d\Sigma = R^2\sin(\theta)\,d\theta d\varphi \;\Rightarrow\; \frac{d\Sigma}{R^2} = \sin(\theta)\,d\theta d\varphi$$

An antenna which strongly prefers to radiate in a given direction will have $f(\vartheta,\varphi)$ close to unity only in a small angle → D_M will be larger! → High directivity=narrow beam

Beamwidth

Directive antennas: most of the radiated power concentrated inside the main beam.
If the power density is constant in a spherical portion with $2\theta_B$ as cone aperture and 0 outside:



$$f(\theta) = \begin{cases} 1 & \text{for } 0 < \theta < \theta_B, \\ 0 & \text{for } \theta_B < \theta < \pi, \end{cases}$$

$$D_M = \frac{4\pi}{\int_0^{\theta_B}\int_0^{2\pi} f(\theta)\sin(\theta)\,d\varphi d\theta} = \frac{2}{1-\cos(\theta_B)}$$

[D_M in linear scale]

The **beamwidth** ($\Delta\theta$) is twice θ_B:

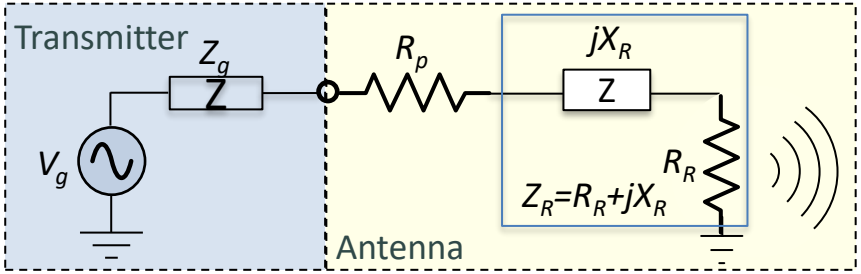
$$\Delta\theta = 2\arccos\left(1 - \frac{2}{D_M}\right)$$

The formula relating Δθ to D_M is a rough but reasonable approximation of HPBW for very directive antennas.
High directivity implies a very narrow beam!

Radiated Power and Electrical Power

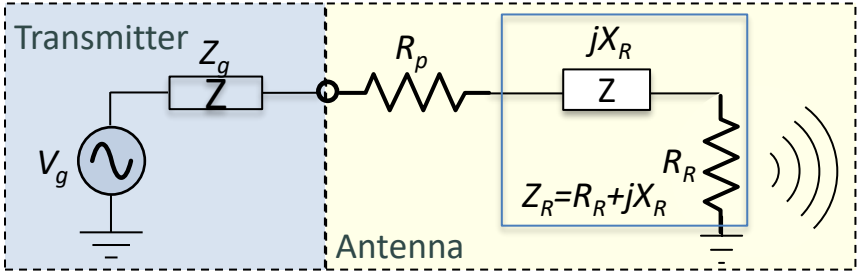
From the transmitter side, the antenna can be represented as an electrical 1-port:

Generator offers P_{av} as available power



The impedance "seen" by the transmitter is determined by two contributions:
Z_R (radiation impedance) and R_p (loss resistance).
Usually, R_p << Real(Z_R)
The power actually transferred to Z_R constitutes the radiated power (P_{rad})

Efficiency



- Power transferred to the antenna: P_T
- Maximum power transfer occurs with conjugate matching:

  $Z_g=(R_p+Z_R)^* \cong Z_R^* \rightarrow P_{av}=P_T$

  (Otherwise, a matching network should be introduced between antenna and transmitter)
- However, even in matching condition, only a fraction of P_T is radiated by the antenna:

$$P_{rad} = P_T \frac{\mathrm{Re}(Z_R)}{\mathrm{Re}(Z_R)+R_p} = \eta P_T$$

η is the *efficiency* of the antenna

Antenna Gain

Radiated power density can be then expressed as function of the power delivered to the antenna:

$$S_R(R,\theta,\phi) = \frac{P_{rad}}{4\pi R^2} D_M f(\theta,\phi) = \frac{P_T}{4\pi R^2} \eta D_M f(\theta,\phi)$$

$$S_R(R,\theta,\phi) = \frac{P_T G}{4\pi R^2} f(\theta,\phi)$$



- $G=\eta D_M$ is the *Maximum Gain* of the antenna (or simply *Gain*)
- G and D_M are normally measured in dBi: $G_{\text{dBi}}=10\text{Log}(G)$; $D_{M\,\text{dBi}}=10\text{Log}(D_M)$
- Effective Isotropic Radiated Power ***(EIRP)***

$$\text{EIRP}=P_T G$$

- It is often measured in logarithmic units (dBm or dBW)
- Regulations often limit the EIRP, rather than the transmitted power

Fields Intensity

For a plane wave propagating in a medium with relative dielectric constant ε_r,

E and H fields intensity determine the power density (S_R):

$$S_R(R,\theta,\phi) = \frac{1}{2}Re(EH^*) = \frac{1}{2}\frac{|E|^2}{Z_W} = \frac{1}{2}Z_W|H|^2 \quad \text{with } Z_W = \frac{377\Omega}{\sqrt{\epsilon_r}}$$ → In air/vacuum: Z_W=377Ω

The power density from a transmitting antenna (P_T, G) at distance R along the direction of maximum radiation is given by: $S_R=P_T\,G/(4\pi R^2)$.
Then, in the direction of maximum radiation:

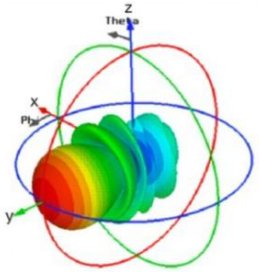
$$|E| = \frac{1}{R}\sqrt{\frac{Z_W EIRP}{2\pi\sqrt{\epsilon_r}}} \qquad |H| = \frac{1}{R}\sqrt{\frac{\sqrt{\epsilon_r}EIRP}{2\pi\, Z_W}}$$

Example: P_T=100 W, G=10 dBi, R=10m → EIRP=1kW (30dBW), |E|=24.5 V/m

Beam efficiency

Definition:

$$BE = \frac{Power\ transmitted\ within\ the\ main\ lobe}{Power\ transmitted\ by\ the\ antenna}$$



Using the previous formulas (assuming the *z* axis as main beam axis):

$$BE = \frac{\int_0^{\theta_B}\int_0^{2\pi} f(\vartheta,\varphi)\sin(\theta)\,d\varphi d\theta}{\int_0^{\pi}\int_0^{2\pi} f(\vartheta,\varphi)\sin(\theta)\,d\varphi d\theta} = \frac{D_M}{D'_M}$$

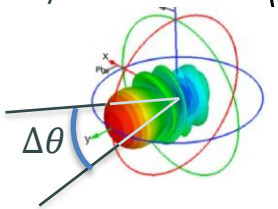
with D'_M the maximum directivity computed assuming that all the power is transmitted in the main lobe

BE measures the capability of confining radiation only in the main lobe, without wasting power in other directions (e.g. secondary lobes)

Transmitting Antennas so far (summary)

- Interface from guided to radiation in free space. Radiates spherical waves always directed away from the antenna
  In far field ($R>d_F$): locally plane waves, E and H fields orthog.: $S_R(R,\vartheta,\varphi)=P_{rad}\, D_M f(\vartheta,\varphi)\,/(4\pi R^2)=|E|^2/(2Z_W)$
- Directivity: radiation preferred in some directions rather than others
- Maximum directivity: $D_M=max\ D(\vartheta,\varphi);\ D(\vartheta,\varphi)=D_M f(\vartheta,\varphi);$
- Antenna impedance: must be conj. matched to Tx internal imped.
  → avail. power from tx (P_{Tx})is entirely transferred to antenna
- Antenna impedance: Radiation impedance + Loss resistance (dissipates a fraction of P_{Tx})
- Gain G: D_M decreased by losses: $G=D_M\eta$ → $S_R(\vartheta,\varphi)=P_{Tx}\, G\, f(\vartheta,\varphi)\,/(4\pi R^2)$
- Radiated characteristics:
  - Half-power Beamwidth (HPBW, Δθ): where $f(\vartheta,\varphi)=0.5$; for directive antennas approximately $\Delta\theta = 2\arccos\left(1-\frac{2}{D_M}\right)$
  - Beam efficiency: capability of confining radiated power into HPBW
  - Side lobe level: difference in dB between main lobe and the highest of secondary lobes
  - Polarization (E field direction): linear or circular, depending on type of antenna
  - Larger antennas are normally more directive, thus have a narrower HPBW

$$D_M = \frac{G}{\eta} = \frac{4\pi}{\int_0^{\pi}\int_0^{2\pi} f(\vartheta,\phi)\sin(\theta)\,d\phi d\theta}$$



Receiving Antenna

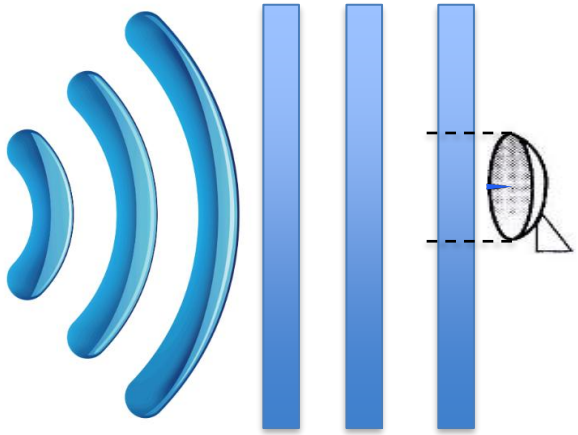
An electromagnetic plane wave incident on a receiving antenna produces an electric power P_r given by :

$$P_r = S_R \cdot A_e \cdot g(\theta, \phi)$$

with S_R power density of the incident wave,

$g(\theta,\phi)$ directivity function (previously called $f(\theta,\phi)$ for the TX antenna) and A_e *effective area* of the receiving antenna

- Intuitively, a receiving antenna "catches" a portion of the wave
- From a physical point of view, the effective area is the ratio between the received power and the power density of the incident wave with the antenna pointed along the maximum of the directivity function.
- The internal electrical dissipation (efficiency factor) is included into A_e.



Relationship between A_e e G

For all antennas it can be shown that A_e, G and the wavelength $\lambda = c/f$ are related:

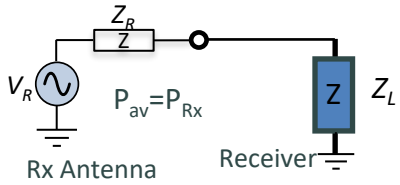
$$\frac{G}{A_e} = \frac{4\pi}{\lambda^2} = \frac{4\pi f^2}{c^2}$$

- The effective area is not equal to the physical area of the antenna (typically is a fraction)

  $A_e = e_a A_{physical}$ (Defined only for directional antennas, like parabolic dishes, horns…)
- Most directional antennas have A_e independent on frequency

  → Gain increases with f^2!

Equivalent circuit of the receiving antenna



If the antenna is pointed in the direction of the incident wave and $Z_L=(Z_{rad})^*$:

$$P_L = P_{r\max} = S_R A_e$$

If the matching condition is not verified, i.e. $Z_L \neq (Z_{rad})^*$:

$$P_L = P_{r,max} \frac{4\,\mathrm{Re}\{Z_R\}\,\mathrm{Re}\{Z_L\}}{|Z_R + Z_L|^2} = k_{mis}^2 P_{r,\max}$$

Mismatch factor k_{mis}^2

More generally, if the antenna polarization is different from the incident wave polarization, the received power $P_{r,max}$ is reduced by a factor k^2_{pol} (=zero if the rx antenna's polarization is orthogonal to the transmitted one!):

$$P_L = k_{mis}^2 \cdot \left(k_{pol}^2 \cdot P_{r,\max}\right) = k_{mis}^2 k_{pol}^2 S_R A_e$$

Example: Dish antenna



Category: Fixed Area

Gain: $$G = A_e \frac{4\pi}{\lambda^2} = A_e \frac{4\pi}{c^2} f^2$$

G increases quadratically with *f*

The effective area A_e is proportional to the dish area:

$$A_e = e_a \frac{1}{4} \pi d^2$$

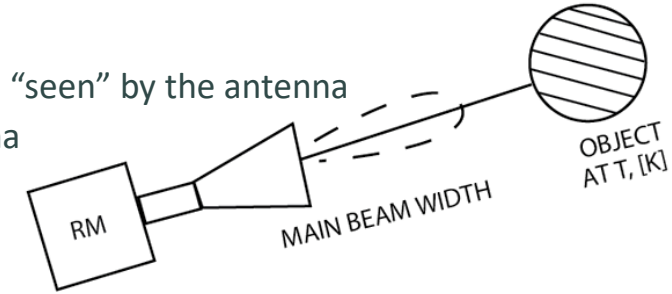
e_a= aperture efficiency (0.55-0.65)
d = Dish diameter
G increases quadratically with *d*

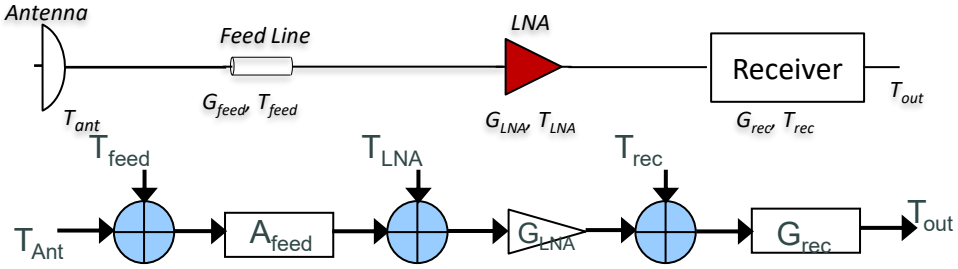
The approximate value of the beamwidth $\Delta\vartheta$ is:

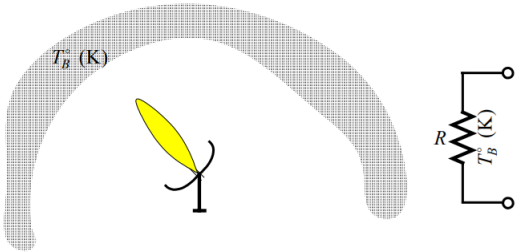
$$\Delta\theta = 2\cos^{-1}\left(1 - \frac{2}{D_{MAX}}\right) = 2\cos^{-1}\left(1 - \frac{2\eta}{G}\right) = 2\cos^{-1}\left(1 - \frac{2\eta\lambda^2}{A_e 4\pi}\right) = 2\cos^{-1}\left(1 - \frac{2\eta}{\pi^2 \cdot e_a}\left(\frac{\lambda}{d}\right)^2\right)$$

Noise at the antenna output

- The receiving antenna gives some noise at its output
- Noise is captured from the intrinsic radiation associated to the bodies "seen" by the antenna
- →Antenna noise is determined primarily by the pointing of the antenna
- Main sources:
  - Clear Sky noise: noise received from the sky with no atmospheric perturbations (but presence of sun affects it)
  - Ground noise: noise radiated by the earth and partially captured (both main and side lobes)
  - Perturbed atmosphere noise: the noise produced by rain, snow, hailstorm, fog, etc.
- Overall generated noise can be represented by an Equivalent Noise Temperature associated to the antenna, which is added to the other equiv. noises: **feed lines (cables/connections, if present) and antenna temp. are in front!**

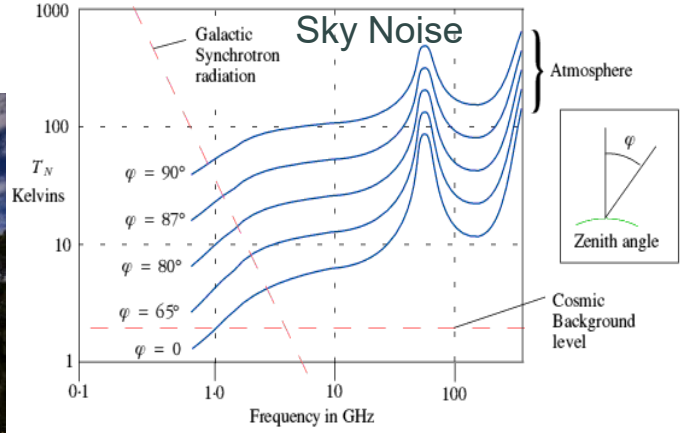


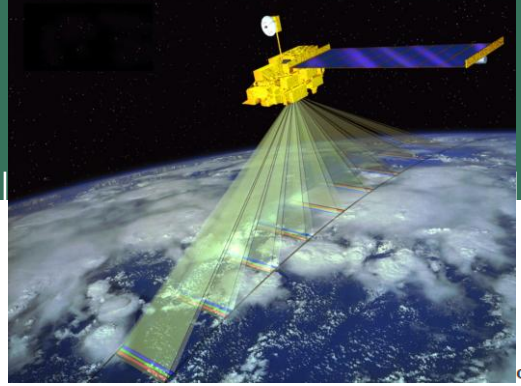




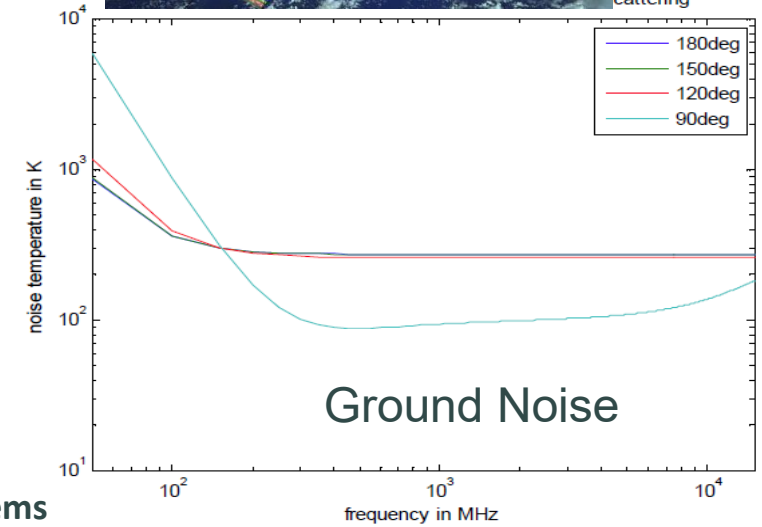
Sky and Ground Noise







cattering



Courses dedicated to this!

EM wave propagation for space-borne systems

Wireless and Mobile Propagation

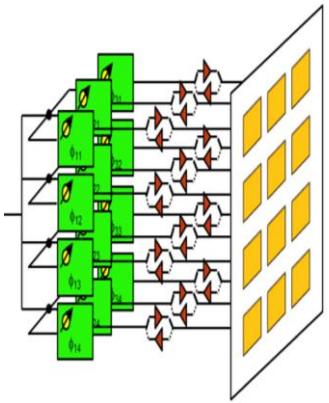
Antenna parameters (summary)

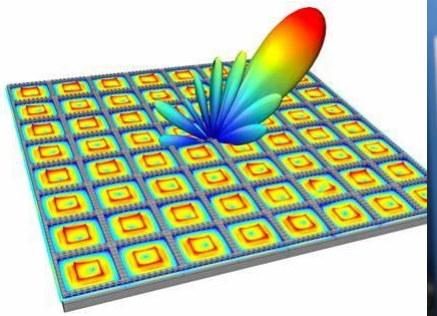
- Gain G (transmission)
- Effective Area A_e (reception)
- Directivity function (same for Tx and Rx): $g(\vartheta,\varphi)=f(\vartheta,\varphi)$
- Antenna Impedance
- Beamwidth and beam efficiency
- Polarization (E field direction)
- Side lobe level
- Equivalent noise temperature T_{ant}

An antenna is “reciprocal” can be used in TX and in Rx with the same gain, efficiency, polarization…

Phased Arrays

- Several small (low directivity) base antennas, each with its own phase delay (adjustable)
- Can be configured to be equivalent to a large directive antenna
- Directivity function can be dynamically modified to point in other directions
- They are also capable to generate a multi-beam directivity function (more than one beam), whose directions can be electronically assigned
- Radar and satellite applications (regional satellite services).
- Mobile access and Fixed Wireless Access
- Their behavior
  can still be described
  by the same parameters
- May additionally have
  “grating lobes”
  (a kind of very high undesired lobe)







The RF Link budget: Friis equation



Transmitter power density (at distance R)

$$S_R(R, \theta, \phi) = \frac{P_t}{4\pi R^2} G_t \cdot f(\theta_t, \phi_t)$$

Received power

$$P_r = S_R \cdot A_e \cdot g(\theta_r, \phi_r)$$

Link Budget (Friis equation):

$$P_r = S_R \cdot A_e \cdot g(\theta_r, \phi_r) = \frac{P_t}{4\pi R^2} G_t \cdot f(\theta_t, \phi_t) \cdot A_e \cdot g(\theta_r, \phi_r)$$

Logarithmic form of Friis Equation

Taking into account the relationship between A_e e G_r:

$$P_r = P_t \left(\frac{\lambda}{4\pi R}\right)^2 G_t \cdot G_r \cdot g(\theta_r, \phi_r) \cdot f(\theta_t, \phi_t)$$

In logarithmic units (dB) :

$$P_{r,dBm} = P_{t,dBm} - 20Log\frac{4\pi R}{\lambda} + G_{t,dBi} + G_{r,dBi} + 10Log(f(\overline{\theta_t}, \overline{\phi_t})) + 10Log(g(\overline{\theta_r}, \overline{\phi_r}))$$

Free space loss (neglects any atmospheric phenomenon such as rain…)

0dB when antennas are perfectly aligned toward each other (reciprocally aligned)

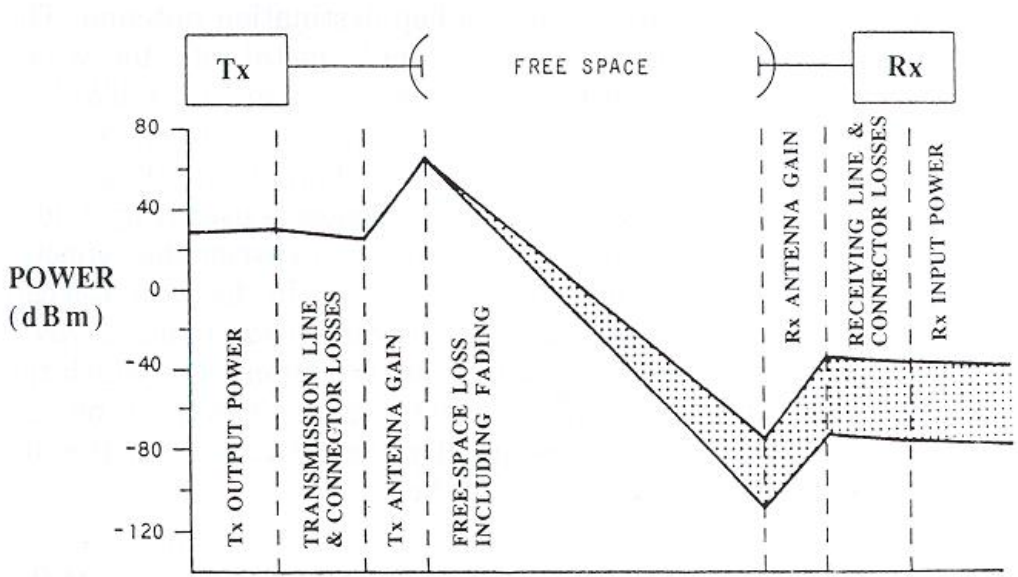
Free space attenuation: $FSL=(4\pi R/\lambda)^2$

In dB: $FSL_{dB}=20\ Log(4\pi R/\lambda)$

$(\bar{\theta}_t, \bar{\phi}_t)$ are the angles at which the transmitting antenna sees the receiving antenna

$(\bar{\theta}_r, \bar{\phi}_r)$ are the angles at which the receiving antenna sees the transmitting antenna

Power Budget Diagram



$P_r = P_t - A_{dt} + G_t - FSL + G_r - A_{dr}$ (all in logarithmic units)

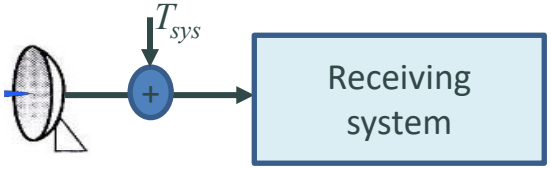
System SNR

A received signal power P_r with band B and assuming the noise is limited to B (with a filter);

All noises are added at the Rx antenna output as an equivalent noise temperature of the system;

SNR of the receiving system is:

$$SNR_{sys} = \frac{P_r}{K \cdot T_{sys} \cdot B}$$



P_r is obtained by the Friis equation:

$$SNR_{sys} = \frac{G_t P_t}{kB}\left(\frac{\lambda}{4\pi R}\right)^2 \frac{G_r}{T_{sys}} = EIRP\,\frac{1}{FSL}\,\frac{1}{kB}\,\frac{G_r}{T_{sys}}$$

Where *FSL* is the free-space loss and the ratio G_r/T_{sys} is an overall figure of merit of the receiving system, including the antenna (it is measured in dB/°K)

→ SNR increases with EIRP: more signal reaches the receiver (more powerful or more directive Tx)

→ SNR decreases with distance (squared): less signal power reaches the receiver

→ SNR decreases with B and T_{sys}: more noise enters together with the signal

→ SNR increases with G_r: receiving antenna captures more signal power

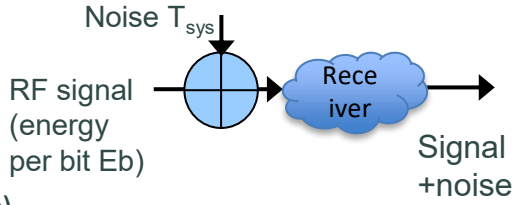
Capacity

- According to Shannon Theorem, the maximum data rate C (bit/s) that can be achieved is (B signal bandwidth in Hz):

$$C = B \log_2(1 + SNR)$$



- Hence, a bit rate R <C is theoretically possible error-free (but it would require an ideal coding scheme with infinite complexity).
- In practice there is a small but not zero probability of error: BER (Bit-error-rate)
- BER is related to the ratio E_b/N_0 (according to the modulation scheme adopted);
- E_b is the energy per bit and $N_0=kT_{sys}$ is the system noise spectral density with Tsys the system noise temperature usually at the input of the receiver (i.e. all noise contributions are gathered in an equivalent single T_{sys} at the input)
- Given a required BER (i.e. 1e-3), a specific modulation and mapping (e.g. 16QAM, Grey) →Eb/N_0 is determined (e.g. 10.5dB)
- The communication system must be designed to guarantee that value of Eb/N0



Data Rate

T_b= time for receiving one bit → R=1/T_b [R=Data rate (bit/s)]
P_r= received power =$E_b/T_b=E_bR$

From the definition of system SNR (antennas reciprocally aligned at the max. of the directivity diagrams):

$$SNR_{sys} = \frac{P_r}{kT_{sys}B} = \left(\frac{E_b}{N_0}\right)\left(\frac{R}{B}\right) = EIRP\,\frac{1}{FSL}\,\frac{1}{kB}\left(\frac{G_r}{T_{sys}}\right)$$

$$\rightarrow \quad R = \frac{EIRP}{E_b/N_0}\,\frac{1}{k\;FSL}\,\frac{G_r}{T_{sys}}$$

R: maximum data rate that can be obtained with a given E_b/N_0 (i.e., with prescribed BER) when G/T of the receiver and transmitted EIRP are also given
Or: if R is given, Gr/Tsys can be computed

Data Rate and Signal Bandwidth

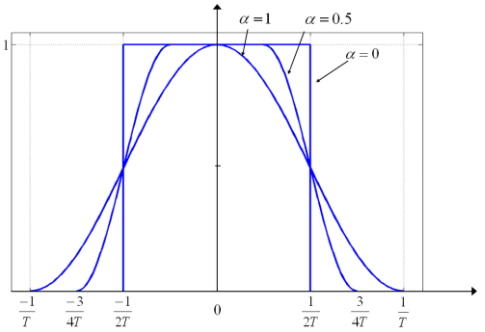
System SNR thus depends on (R/B) ratio: $SNR_{sys} = \left(\frac{E_b}{N_0}\right)\left(\frac{R}{B}\right)$

Eb/N0 however is imposed by the BER and modulation scheme adopted.

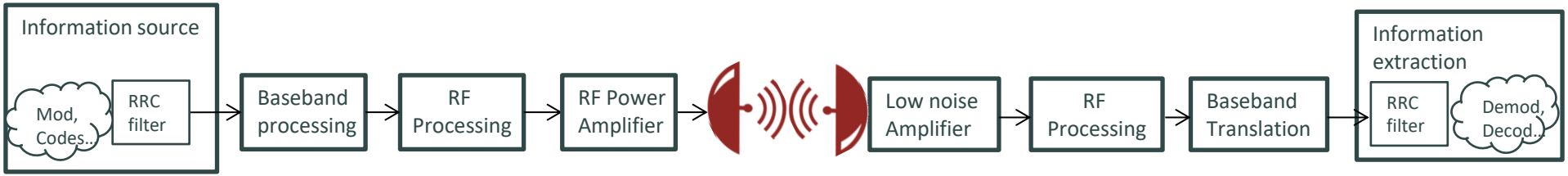
Moreover, in case of M-QAM, B and R are related:

$$B = \frac{R}{\log_2 M} \cdot (1 + \alpha)$$

α is the roll-off factor of the root-raised-cosine filter (RRC filter)



After computing R (formula in previous slide, or from requirements), B or M can be computed



Map of Relations: every node is a relation

If all but one quantities entering a node are known, the unknown can be computed

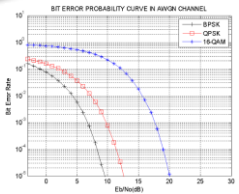
BER

Modulation Scheme *M*

Rolloff α

$B = \frac{R}{\log_2 M} \cdot (1 + \alpha)$

Rate R



$SNR_{sys} = \left(\frac{E_b}{N_0}\right)\left(\frac{R}{B}\right)$

B

$SNR_{sys} = \frac{P_r}{kT_{sys}B}$

Eb/N0

SNRsys

Noise T_{sys}

EIRP, G_r, FSL

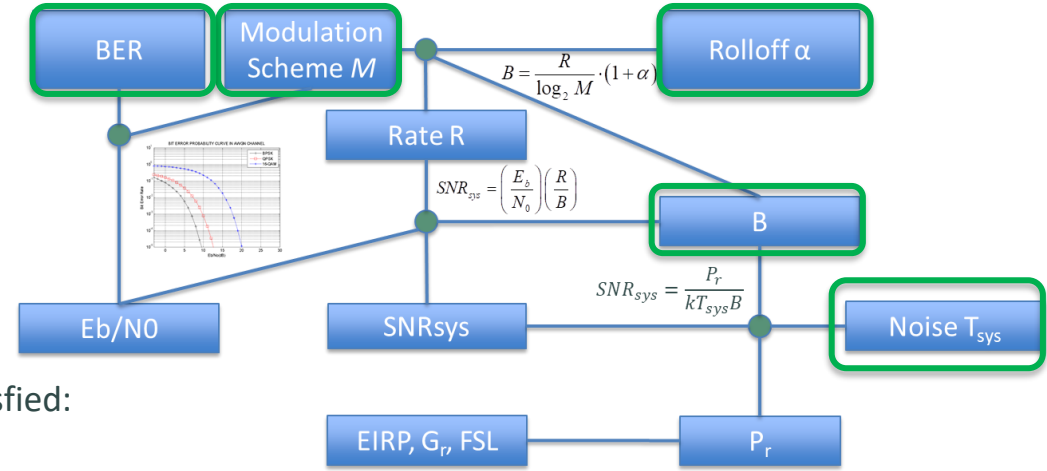
P_r

Example: needed received power (typical real world scenario)

Given: B, modulation, BER, T_{sys} → Find P_r

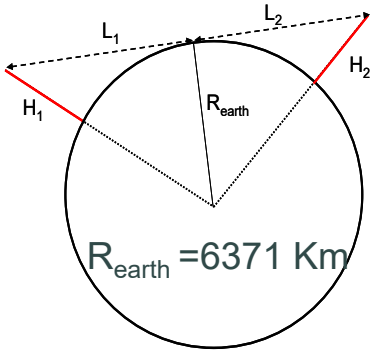
The channel bandwidth is given by the regulations: B=56MHz, f=18GHz
BER is required by application: $BER=10^{-3}$
The system parameters are: Rolloff=0.25, T_{sys}=8600K

1. Data rate is determined by:
   $$R = \frac{B}{(1+\alpha)} \cdot \log_2 M$$ →R=179Mbit/s
2. From desired BER, E_b/N_0 is found
   (using curves of the chosen modulation)
   →Eb/N0=17.5dB
3. The required SNR_{sys} can then be computed :
   $$SNR_{sys} = \left(\frac{E_b}{N_0}\right)\left(\frac{R}{B}\right)$$ → SNR_{sys}=180=22.6dB
4. Finally, P_r is found so that SNR equation is satisfied:
   $P_r = k\, T_{sys}\, B\, SNR_{sys}$ = 6.64pW (-82dBm)



From this, if needed, one can compute EIRP or G_r or distance…

Terrestrial Radio Link: maximum length



The maximum length is limited by the Earth's surface curvature.

Simplified conditions:
- Earth is a sphere
- Wave scattering produced by the atmosphere is negligible (no "extended radio horizon")

Maximum link distance is determined by the visibility condition of two points placed at heights H_1 and H_2:

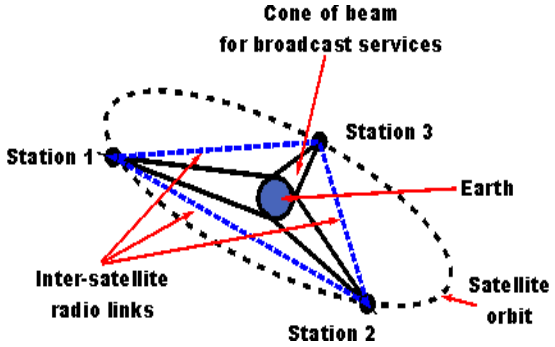
$$L_1 = \sqrt{(H_1 + R_{earth})^2 - R_{earth}^2} = \sqrt{H_1^2 + 2H_1R_{earth}} \sim \sqrt{2R_{earth}H_1} \qquad L_2 = \sim\sqrt{2R_{earth}H_2}$$

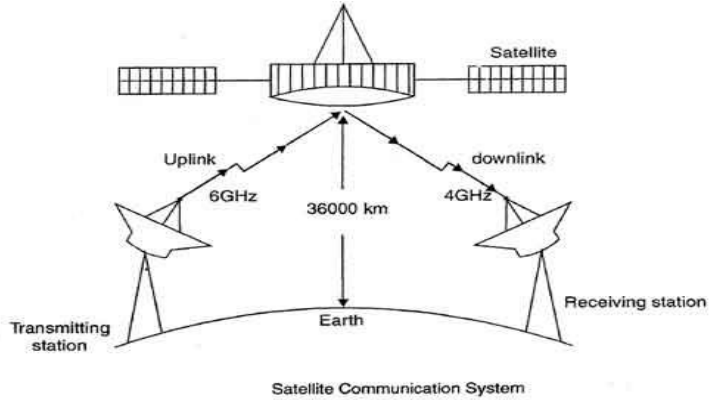
$$L_{max} = L_1 + L_2 \sim \sqrt{2R_{earth}}\left(\sqrt{H_1} + \sqrt{H_2}\right) = 3.57km\left(\sqrt{\frac{H_1}{1m}} + \sqrt{\frac{H_2}{1m}}\right)$$

Example: Tower 1: 10m;
Tower 2: 30m → Lmax=30.8km

Satellite Communications

3 geostationary satellites are sufficient to cover the whole Earth's surface:



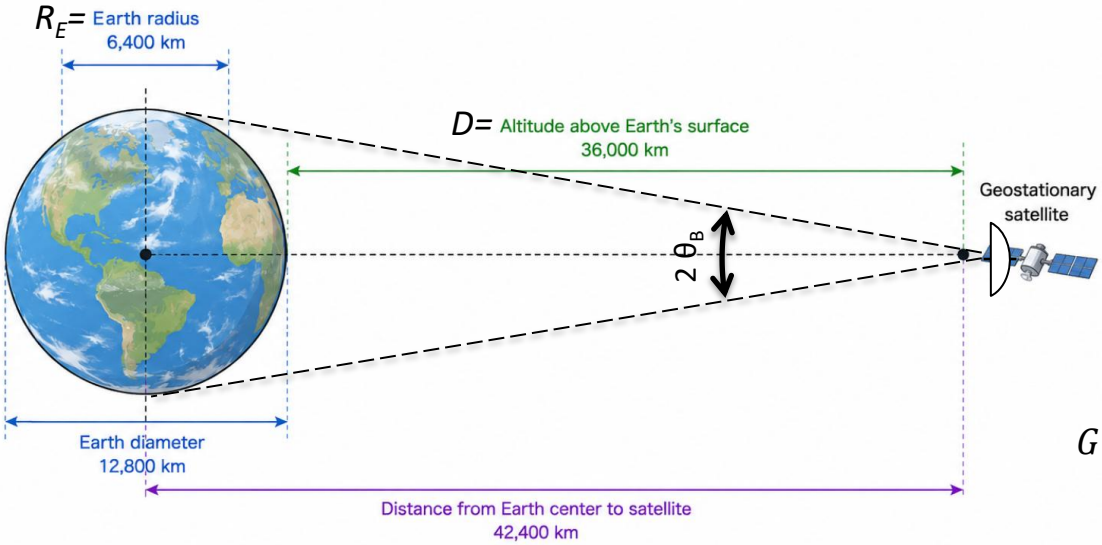


Satellite Communication System

A geostationary satellite rotates in the same direction as the rotation of the Earth and has an approximate 24 hour period. This means that it remains in the same position relative to the Earth.
Examples of loss in satellite link:

FSL (@6GHz)~20Log(4πd/λ)=199dB! FSL (@4GHz)~20Log(4πd/λ)=195.6dB!

Beamwidth of satellite antennas



θ_B is obtained by imposing the distance of geo satellite (D=~36000 Km) and the earth's radius (R_E=6371 Km):

$$2\theta_B = 2\arctan\frac{R_E}{D+R_E} = 17.36°$$

Antenna (assume a lossless dish with η=1) must exhibit:

$$G = \left(\frac{2}{1-\cos(\theta_B)}\right) = 174.62\ (\sim 22.4\ \text{dB}),$$

$$d = \frac{\lambda}{\pi}\sqrt{\frac{G}{e_a}} = 27.15\ \text{cm} \quad (f{=}6\ \text{GHz} \rightarrow \lambda{=}5\text{cm}, e_a = 0.6)$$

Antennas for Earth Stations

They must exhibit a very large gain (EIRP from earth station is extremely high)

Dish antennas with a very large diameter: 10-30m

Gain up to 60 dB → very narrow beam (fraction of degree)

To maintain the correct pointing of the satellite an automatic system is necessary (a beacon signal transmitted from the satellite, usually)

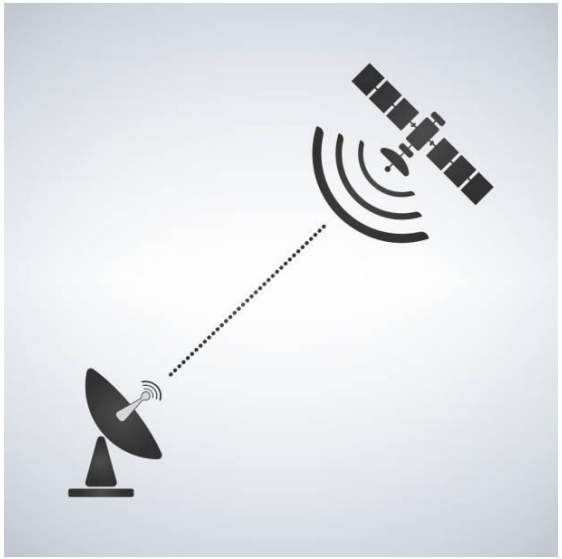
Example:

f=6GHz, diameter=20m → Aphysical=314m^2

Assume e_a=0.8 → Ae~251m^2 → Gain G=1.26e6 (61dBi)

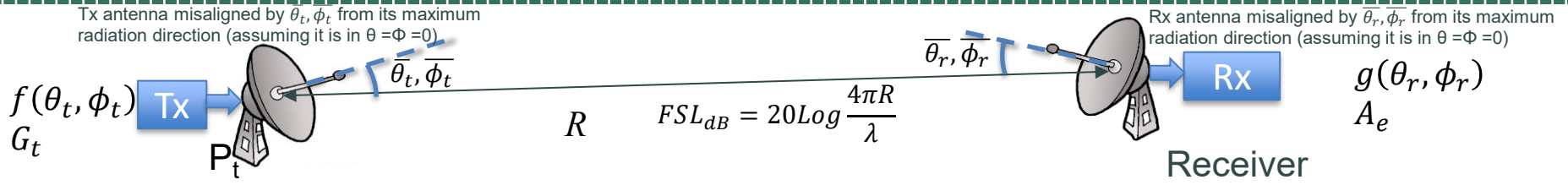
Assume η=0.7→D_M=G/η=1.8e6 (62.6dBi)

HPBW: Δθ=0.17°



$$\Delta\theta = 2\cos^{-1}\left(1-\frac{2}{D_M}\right) = 2\cos^{-1}\left(1-\frac{2\eta}{\pi^2\cdot e_a}\left(\frac{\lambda}{d}\right)^2\right)$$

Recap of Link Budget
(neglecting polarization loss and mismatch loss)



$$P_{r,dBm} = P_{t,dBm} - FSL_{dB} + G_{t,dBi} + G_{r,dBi} + 10Log(f(\overline{\theta_t}, \overline{\phi_t})) + 10Log(g(\overline{\theta_r}, \overline{\phi_r}))$$

E_b/N_0: determined from modulation (e.g. M-QAM) and required BER (e.g. 10^{-3})

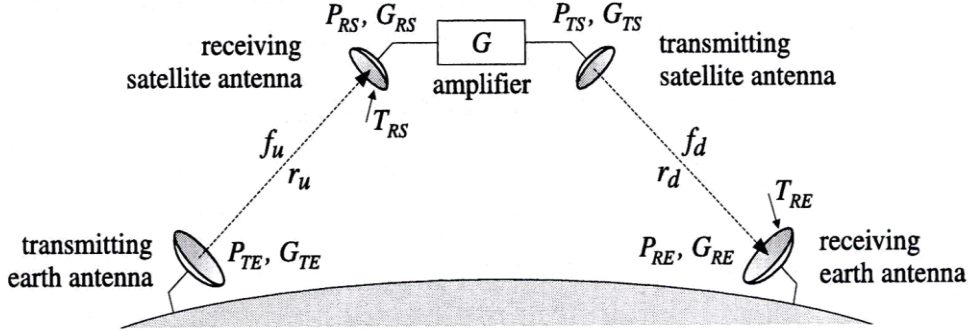
R/B determined by modulation and RRC filter rolloff α: $R/B=(1+\alpha)/\log_2 M$

SNR and E_b/N_0 related by R and B: SNR= (E_b/N_0) (R/B)

Fulfilling a required SNR thus involves:

- Align reciprocally the antennas (Rotate the antennas so that $f(\overline{\theta_t}, \overline{\phi_t})$ and $g(\overline{\theta_r}, \overline{\phi_r})$=1, assuming the polarization axes are already aligned, otherwise a further factor -20Log(k_{Pol}) would appear)
- Improve the G_r/T_{sys} of the receiver (reduce system noise temperature of the receiver T_{sys} or use a more directive Rx antenna)
- Increase the EIRP of the transmitter (increase Tx power or use a more directive Tx antenna)

Satellite Link as a Repeater



f_u=Up-link frequency (6 GHz), f_d=Down-link frequency (4 GHz)
r_u, r_d>36000 Km (depends on the angle beneath which the satellite is seen)

G_{TE}, G_{RS}, G_{TS}, G_{RE}= Gain of earth and satellite antennas
P_{TE}, P_{TS}=Transmitting powers (earth and satellite)
G = Gain of satellite transponder

Friis Equation (best pointing of the antennas):

$$P_{RE} = P_{TE} \cdot G_{TE} \cdot G \cdot G_{RS} \cdot G_{TS} \cdot G_{RE} \cdot \left(\frac{\lambda_u}{4\pi r_u}\right)^2 \cdot \left(\frac{\lambda_d}{4\pi r_d}\right)^2$$

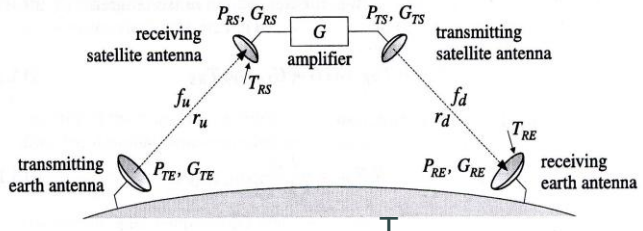
SNR of a Radio Link with a Repeater

T_{RS}= Noise equivalent temperature of the satellite receiving system

T_{RE}= Noise equivalent temperature of the earth receiving station

$$SNR_u = \frac{P_{RS}}{kT_{RS}B} \quad \text{(Upper link)}$$

$$P_{RS} = P_{TE} \cdot G_{TE}G_{RS} \cdot \left(\frac{\lambda_u}{4\pi r_u}\right)^2$$



$$SNR_d = \frac{P_{RE}}{kT_{RE}B} \quad \text{(Down link)}$$

$$P_{RE} = P_{RS} \cdot G \cdot G_{TS}G_{RE} \cdot \left(\frac{\lambda_d}{4\pi r_d}\right)^2$$



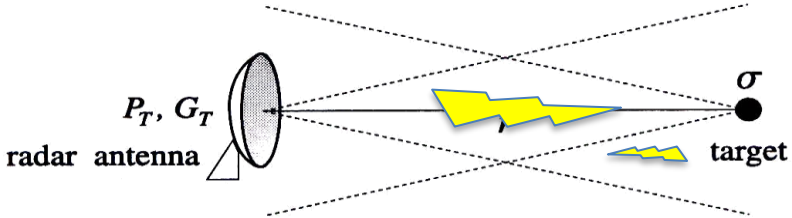
$$T_{RT} = T_{RE} + T_{RS} \cdot G \cdot G_{TS}G_{RE} \cdot \left(\frac{\lambda_d}{4\pi r_d}\right)^2 = T_{RE} + T_{RS} \cdot \frac{P_{RE}}{P_{RS}}$$

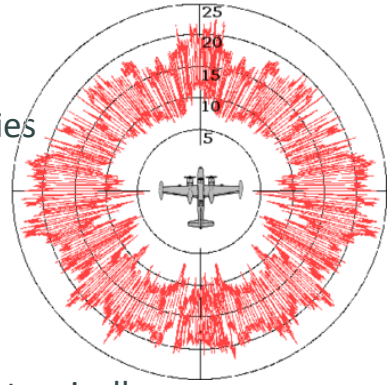
$$SNR_{tot} = \frac{P_{RE}}{kT_{RT}B} = \frac{P_{RE}}{k\left(T_{RE} + T_{RS} \cdot \frac{P_{RE}}{P_{RS}}\right)B} = \frac{1}{\frac{kT_{RE}B}{P_{RE}} + \frac{kT_{RS}B}{P_{RS}}}$$

$$SNR_{tot} = \frac{1}{\frac{1}{SNR_u} + \frac{1}{SNR_d}}$$

Monostatic Radar Equation

Radar in TX: radiates a strong power in a given direction





Incident wave: impinges on target→ reflected back (retransmitted) by metals and discontinuities

Retransmitted wave: propagates and is intercepted by the radar antenna

Distance to target: estimated by delay or by correlation methods (pulsed or Continuous Wave)

The concept of *radar cross section* σ provides a measure of the effective area of the target:
σ is measured as an area (m^2)

$$\sigma = \frac{P_{\text{target}}}{S_T}$$

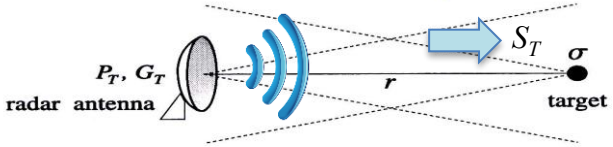
P_{target}: Power reflected by the target, assumed isotropically
S_T=Power density incident on the target

If the target scatters directionally toward the radar (e.g. orthogonal reflecting plane) → high RCS in that direction
If instead it scatters the impinging power away in all directions → low RCS
Monostatic RCS is a function of the impinging direction (θ, ϕ) in general

Monostatic Radar Budget

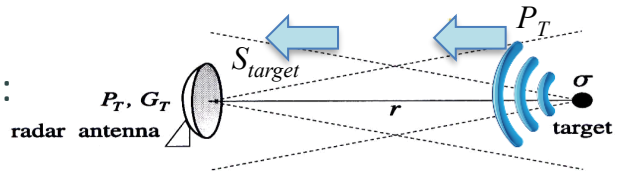
Assuming the antenna pointed for the maximum of directivity, we have at the target position:

$$S_T = \frac{P_T G_T}{4\pi r^2} \qquad P_{target} = \sigma S_T = \sigma \frac{P_T G_T}{4\pi r^2}$$



The power density of the wave reflected back to the radar position:

$$S_{target} = \frac{P_{\text{target}}}{4\pi r^2} = \sigma \frac{P_T G_T}{(4\pi r^2)^2}$$

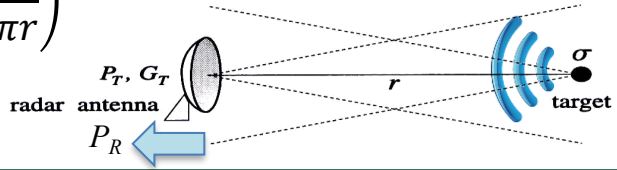


Finally, the power at the radar receiver is given by the effective area A_R of the radar antenna:

$$P_R = A_R S_{target} = \sigma \frac{P_T G_T A_R}{(4\pi r^2)^2} = \sigma \frac{P_T G_T^2 \lambda^2}{(4\pi)^3 r^4} = P_T \cdot G_T^2 \cdot \frac{4\pi\sigma}{\lambda^2} \cdot \left(\frac{\lambda}{4\pi r}\right)^4$$

$$A_R = \frac{G_T \ \lambda^2}{4\pi}$$

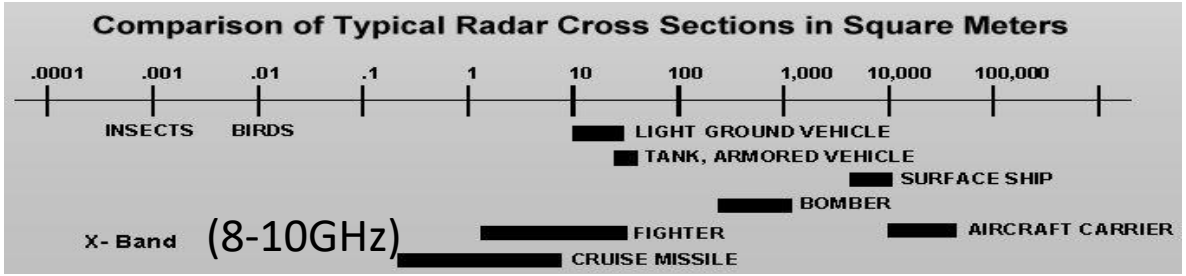
$$P_{R,dBm} = P_{T,dBm} + 2G_{T,dB} + 10\log_{10}\left(\frac{4\pi\sigma}{\lambda^2}\right) - 2 \cdot FSL_{dB}$$



Monostatic Radar Budget

The minimum detectable received power defines the maximum distance at which the target can be detected:

$$P_{R,min} = \sigma \frac{P_T G_T A_R}{(4\pi)^2 R_{\max}^4} \rightarrow R_{max} = \left[\frac{\sigma P_T G_T A_R}{(4\pi)^2 P_{R,\min}} \right]^{\frac{1}{4}}$$



$P_{r\ min}$ must be sufficiently above noise

→ The radar receiver must be very low noise and very narrow band

→ Antenna directivity helps!